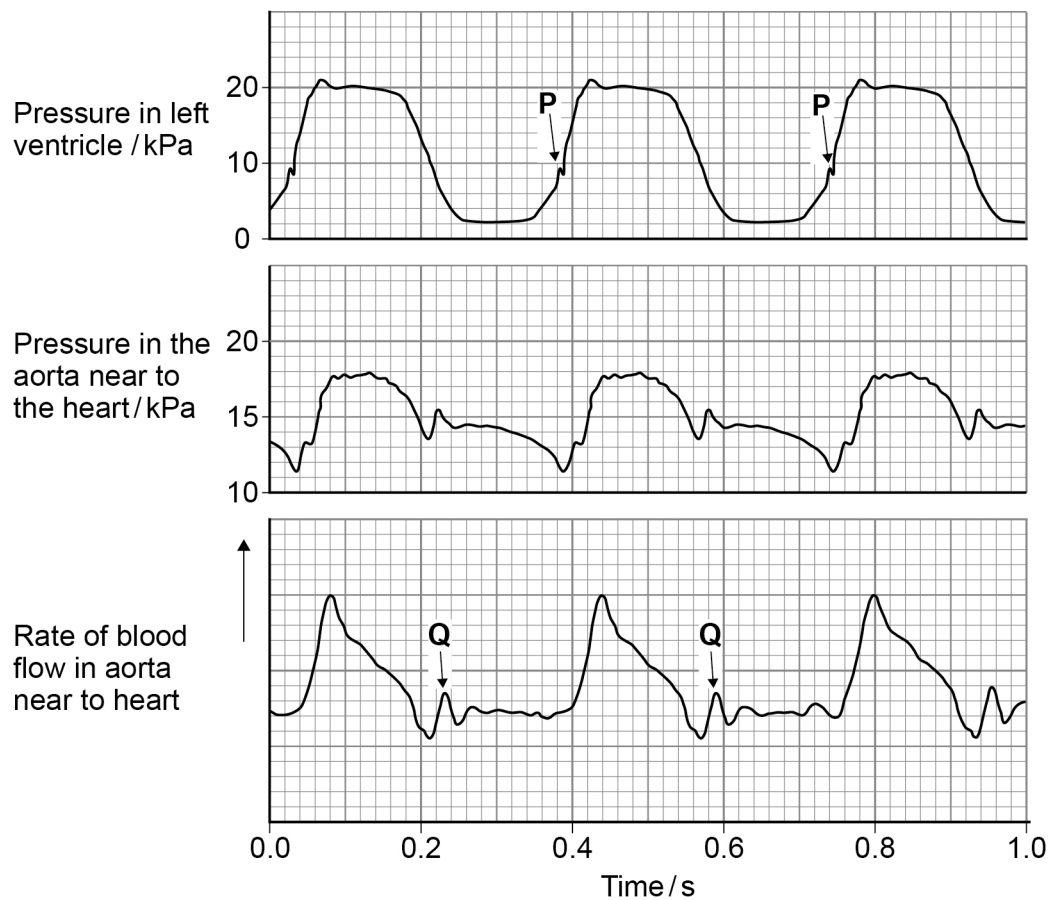


0 3

Figure 3 shows pressure and blood flow during the cardiac cycle in a dog.

Figure 3



0 3 . 1

At **P** on **Figure 3**, the pressure in the left ventricle is increasing. At this time, the rate of blood flow has not yet started to increase in the aorta.

Use evidence from **Figure 3** to explain why.

[2 marks]

0 3 . 2

At **Q** on **Figure 3** there is a small increase in pressure **and** in rate of blood flow in the aorta.

Explain how this happens **and** its importance.

[2 marks]

0 3 . 3

A student correctly plotted the right ventricle pressure on the same grid as the left ventricle pressure in **Figure 3**.

Describe **one** way in which the student's curve would be similar to and **one** way it would be different from the curve shown in **Figure 3**.

[2 marks]

Similarity _____

Difference _____

0 3 . 4

Use information from **Figure 3** to calculate the heart rate of this dog.

[1 mark]

Heart rate _____ beats minute⁻¹

7

Turn over ►
